

- 15 Succinic dehydrogenase is an enzyme that catalyses the conversion of succinate to fumarate in aerobic respiration.

Malonate is a reversible inhibitor of succinic dehydrogenase. Malonate reduces the enzyme's activity by binding to its active site. Malonate and succinate **cannot** bind to the active site at the same time.

Which statement describes the effect of malonate on the activity of succinic dehydrogenase?

- A In the presence of malonate, $V_{\max}$ can still be reached if the concentration of succinate is increased.
- B Malonate has no effect on the K_m .
- C In the presence of malonate, $V_{\max}$ can still be reached if the concentration of fumarate is increased.
- D Malonate decreases the K_m .